

Algebra II with Trigonometry

Chapter 11.4

Olympiad Problems (Gemini)

Saved tutoring response with MathJax rendering. Original PDF file:

[4672988_alg-2trig-h-chp-11-4-note-docx-1.pdf](#)

Problems

1. An ellipse in the coordinate plane has its foci at $(3, 4)$ and $(-5, 12)$. If the ellipse passes through the origin, what is the length of its major axis?
2. The graph of the equation $x^2 - y^2 - 4x + 6y - 5 = 0$ and the line $x = 10$ enclose a bounded region in the coordinate plane. What is the area of this region?
3. A parabola has its focus at $(0, 7)$ and its directrix is the horizontal line $y = -7$. If a point P on this parabola is at a distance of 25 from the focus, what is the shortest distance from P to the x -axis?
4. Consider the hyperbola given by the equation $3x^2 - y^2 - 12x + 10y - 100 = 0$. What is the degree measure of the acute angle between its asymptotes?
5. An ellipse has its foci at $(3, -4)$ and $(3, 4)$, and it passes through the origin. What is the maximum y -coordinate of any point on this ellipse?
6. For what real value of k does the graph of the equation $x^2 + 4y^2 - 10x + 24y + k = 0$ degenerate into a single point?